

Abhirup Choudhury

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PROFESSIONAL SUMMARY

Data Engineer with 1.5+ years of experience building cloud-native ETL pipelines, real-time streaming systems, and scalable data lake architectures on AWS and GCP. Skilled in pipeline orchestration with Apache Airflow, data transformation using DBT and PySpark, and serverless computing with Lambda and Cloud Functions. Proficient in BigQuery, Snowflake, PostgreSQL, and data warehousing. Strong collaborator with cross-functional teams including data scientists and analysts. GATE 2025 qualified in Data Science & Artificial Intelligence.

WORK EXPERIENCE

SBSF Agritech

Jan. 2026 – Present · 3 months

Data Engineer Intern

Hybrid

- Architected a scalable climate data lake on **AWS S3**, automating ingestion of ERA5-Land reanalysis data across **14 environmental features** covering multi-year historical datasets using **Python** and **Bash**.
- Developed event-driven Extract, Transform, Load (ETL) workflows using **AWS Lambda** and **EC2** to retrieve, decompress, and process GRIB files from the Copernicus CDS API, eliminating **100%** of manual data preparation.
- Reduced cloud storage costs by **60%** through **S3 Glacier Deep Archive** lifecycle policies and serverless aggregation logic compared to standard storage tiers.
- Engineered idempotent download logic with structured error handling, logging, and **CloudWatch** monitoring, achieving **zero data loss** across all pipeline executions.
- Designed dimensional data models and schema definitions for structured climate storage, enabling efficient **SQL** querying and downstream analytical workflows.

Earthmetry Decision Systems

Dec. 2024 – Present · 1 year 4 months

Product Engineer & Data Engineer Intern

Hybrid – Chennai, TN

- Built a live **AQI** monitoring dashboard for the Brihanmumbai Municipal Corporation (BMC), processing real-time streaming data from **2,200+ IoT sensors** with sub-minute latency using event-driven architecture on **GCP**.
- Designed a serverless ingestion pipeline using **Cloud Functions**, **Cloud Run**, and **Cloud Scheduler** for automated **BigQuery** loading, supporting continuous ELT workflows across multiple sensor streams.
- Created a custom geo-batching algorithm to bypass Google Maps API's 20-location constraint, enabling retrieval of **1,000+ coordinate points** per request for large-scale spatial analysis.
- Integrated **Folium** for geospatial visualization with automated ROI mapping using **500m radius buffers**, generating coverage maps across cities including Surat and Bhubaneswar.
- Engineered high-throughput collection workflows processing **10,000+ JSON records/hour** with **99%** validation accuracy, ensuring data quality and integrity for downstream analytics.
- Optimized transformation workflows using **Apache Airflow** for orchestration and **DBT** for modeling, reducing processing time by **40%** and delivering structured datasets on automated CI/CD schedules.

PROJECTS

Gym Performance Optimizer | *Python, MediaPipe, ESP32, TensorFlow, Sensor Fusion, IoT*

- Built an end-to-end fitness monitoring system using sensor fusion combining accelerometer, gyroscope, and IMU data from **ESP32** microcontrollers to analyze gym performance in real time.
- Implemented AI-driven posture detection using **MediaPipe Pose** achieving real-time form analysis at **30 FPS** with multi-modal sensor input processing.

Medical Chatbot | *Python, NLP, HuggingFace Transformers, TensorFlow, RAG, Vector Database*

- Developed a Retrieval-Augmented Generation (RAG) medical query bot ingesting structured documents from the Gale Encyclopedia of Medicine using NLP-based chunking and vector retrieval.
- Leveraged **HuggingFace Transformers** for NLU-based answer extraction, improving query resolution accuracy over keyword-based baselines.

Text-to-Image Generation Pipeline | *Python, HuggingFace Diffusers, Stable Diffusion, CUDA*

- Built a generative AI pipeline using **Stable Diffusion** (dreamlike-diffusion-1.0 and SDXL) via **HuggingFace Diffusers**, enabling prompt-based image synthesis with configurable inference parameters.
 - Optimized inference using **torch.float16** precision on **CUDA GPU**, reducing generation time while supporting multi-image batch outputs.
- Gesture-Based Control System** | *Python, OpenCV, MediaPipe, Real-Time Processing*
- Designed a real-time hand gesture recognition system achieving **90% classification accuracy** for laptop control (volume, brightness) using **MediaPipe Hands** landmarks.
 - Optimized the inference pipeline to achieve **sub-1 second response latency** end-to-end, enabling seamless touchless interaction.

EDUCATION

IIITDM Kancheepuram	2022 – 2026
<i>B.Tech in Smart Manufacturing</i>	CGPA: 7.9/10
IIT Madras BS Degree	2023 – Present
<i>Diploma in Data Science</i>	CGPA: 8.10/10

TECHNICAL SKILLS

Languages: Python, SQL, Java, Scala, C, Bash, HTML/CSS

Big Data & Processing: Apache Spark, PySpark, Apache Kafka, Hadoop, Apache Airflow, DBT

Cloud Platforms: AWS (Lambda, S3, EC2, Redshift, Glacier, IAM, CloudWatch), GCP (Cloud Run, Cloud Functions, Cloud Scheduler, BigQuery, Pub/Sub)

Databases & Warehousing: Snowflake, PostgreSQL, MySQL, MongoDB, Data Modeling, Dimensional Modeling

DevOps & Tools: Docker, Kubernetes, Git, GitHub Actions, Terraform, CI/CD, REST APIs, FastAPI, Postman

Data Science & ML: TensorFlow, OpenCV, MediaPipe, Streamlit, Folium, Pandas, NumPy

CERTIFICATIONS & ACHIEVEMENTS

- Qualified **GATE 2025** in **Data Science & Artificial Intelligence** (DA) — while in the third year of B.Tech.
- Achieved **96 percentile** in JEE Mains 2022 among 1M+ candidates.